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RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Mechatronics Engineering

Assignment On

5-Bar Planar Parallel Robot for Automated Chocolate Tray Packaging

Submitted By:

Md. Ahsan Habib
ID: 2008032
4th Year, Even Semester
Department of Mechatronics Engineering
Rajshahi University of Engineering &
Technology, Rajshahi-6204.

Submitted To:

Dr. Dip Kumar Saha
Assistant Professor
Department of Mechatronics Engineering
Rajshahi University of Engineering &
Technology, Rajshahi-6204.

Md. Robiul Islam
Assistant Professor
Department of Mechatronics Engineering
Rajshahi University of Engineering &
Technology, Rajshahi-6204.

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**MTE 4205
Robotics**

ASSIGNMENT TITLE

5-Bar Planar Parallel Robot for Automated Chocolate Tray Packaging

1. Kinematic Design

A 5-bar planar parallel manipulator is selected as the robotic platform. It consists of two independently driven revolute joints mounted on a fixed horizontal base, each powering a two-link open chain whose distal ends are pin-joined at a common end-effector point. This closed-chain architecture provides high stiffness, low moving inertia, and a large dexterous workspace, all of which are essential for a fast and precise chocolate pick-and-place task.

1.1 Architecture Overview

- Type: 5-bar planar parallel robot (symmetric, 2-DOF)
- Active joints: q_1 (left motor at O_1) and q_2 (right motor at O_2)
- Passive joints: A (junction of L_1 and L_2) and B (junction of L_3 and L_4)
- End-effector: Point P shared distal tip of links L_2 and L_4
- Motion plane: Horizontal XY plane; vertical (Z) motion handled by end-effector actuator
- Degrees of freedom: 2 (sufficient for full X-Y coverage of the tray)

1.2 Coordinate Frame and Design Parameters

The coordinate frame is defined with the X-axis pointing toward the tray and the Y-axis running along the tray width. The base joints O_1 and O_2 are offset to the left of the tray to accommodate the conveyor pick-up area.

Parameter	Value	Justification
$L_1 = L_3$ (proximal links)	20 cm	Reach to far tray corner $\approx 37.4 \text{ cm} < 40 \text{ cm}$ ($L_1 + L_2$)
$L_2 = L_4$ (distal links)	20 cm	Symmetric; maximises internal workspace area
O_1 (left base joint)	$[-8, 10] \text{ cm}$	Positioned left of conveyor; centred vertically on tray
O_2 (right base joint)	$[-2, 10] \text{ cm}$	6 cm right of O_1 ; both motors on the mounting platform
Base separation $ O_1O_2 $	6 cm	Compact mount; all 100 positions verified reachable
Pick-up point (conveyor)	$[4, 10] \text{ cm}$	Centre of conveyor pick-up region
Tray region	$X \in [8, 28],$ $Y \in [0, 20] \text{ cm}$	20×20 cm tray to the right of the conveyor

Parameter	Value	Justification
Chocolate grid spacing	$20/9 \approx 2.22$ cm	10 dimples across 20 cm $\rightarrow$ 9 equal gaps

1.3 Workspace Verification

The reachability of every grid point was verified analytically. For the symmetric design with $L_1 = L_2 = 20$ cm, the maximum reach per chain is 40 cm. Critical tray corners give:

- Far corner [28, 20]: $|O_1P| \approx 37.4$ cm < 40 cm ✓
- Far corner [28, 20]: $|O_2P| \approx 31.6$ cm < 40 cm ✓
- Near corner [8, 0]: $|O_1P| \approx 18.9$ cm < 40 cm ✓
- Pick-up point [4, 10]: $|O_1P| = 12$ cm and $|O_2P| = 6$ cm ✓

All 100 dimple positions and the conveyor pick-up point lie within the simultaneous reachable workspace of both chains.

1.4 Sensor and Vision System

- Overhead RGB camera (e.g., Basler acA1920-40uc) mounted 500 mm above the conveyor; detects chocolate colour and centroid position in real time.
- Conveyor incremental encoder: synchronises pick timing with belt speed.
- Vacuum sensor: confirms successful grasp before the robot begins the place trajectory.

2. 3D CAD Model

The 3D CAD model of the 5-bar parallel robot was developed in SolidWorks. Figure 1 presents the complete assembly, showing the motor housing block, the four hollow-section links, the passive joint connectors, and the pneumatic end-effector actuator. The model confirms that all components fit within the intended mounting envelope and that the links do not collide throughout the operational workspace.

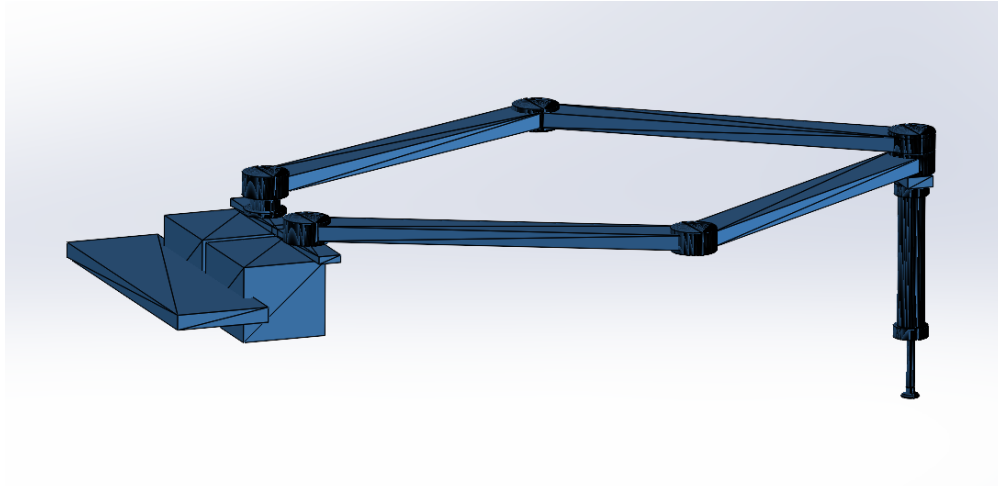


Figure 1. SolidWorks 3D CAD model of the 5-bar planar parallel robot assembly showing the motor housing block (left), proximal and distal links, passive joint connectors, and the pneumatic end-effector with piston rod (right).

2.1 Motor Housing Block

The left-hand assembly visible in Figure 1 is the motor housing block, which serves as the structural base of the robot. It is a solid aluminium alloy box machined to accommodate two servo motors mounted in a stacked arrangement at positions O_1 and O_2 . The two motor shafts protrude from the front face of the block and are aligned horizontally, separated by 60 mm (the base separation d). The block is bolted to the mounting platform through four counterbored holes on its underside. The compact stacked layout keeps the two active joint axes close together, which positions the robot's dexterous workspace directly over the conveyor and tray region. The motor housing also integrates the servo drive electronics and cable routing channels on its rear face.

2.2 Proximal Links (L_1 and L_3)

The two proximal links are identical hollow rectangular-section beams, each 200 mm long and manufactured from aluminium alloy. As seen in Figure 1, they extend outward from the motor shafts at O_1 and O_2 respectively and are driven directly by the servo motors via rigid couplings. The hollow cross-section provides the required bending stiffness while minimising the moving mass, which is critical for achieving high pick-and-place cycle rates. A precision deep-groove ball bearing is pressed into each proximal end to interface with the motor shaft, and a second bearing at the distal end forms the passive revolute joint connecting to the corresponding distal link.

2.3 Distal Links (L_2 and L_4)

The two distal links are identical to the proximal links in cross-section and material, also 200 mm long. In Figure 1, the distal links are visible as the second pair of beams that run from the passive joint connectors (at the top of the linkage assembly) toward the end-effector on the right. Both distal links share a common revolute pin joint at their far ends, which constitutes the end-effector point P. The closed-chain constraint means that whenever the two servo motors command joint angles q_1 and q_2 , the four links collectively determine a unique position for point P, which is exploited by the kinematic model described in Chapter 4.

2.4 Passive Joint Connectors

At the corners of the linkage assembly, cylindrical joint connectors are visible in Figure 1. These are the passive joints A (connecting L_1 and L_2) and B (connecting L_3 and L_4), as well as the shared end-effector joint P (connecting L_2 and L_4). Each connector is a precision-machined aluminium hub housing a deep-groove ball bearing. The bearing inner race is press-fitted onto a hardened steel pin, and the outer race is retained in the hub bore with a circlip. This design eliminates backlash and provides smooth, low-friction rotation at each passive joint, which is essential for kinematic accuracy. The shared connector at P also carries the end-effector mounting thread for attaching the pneumatic cylinder assembly.

2.5 End-Effector Actuator

On the far right of Figure 1, a vertical cylindrical rod is visible extending downward from the end-effector joint connector. This is the piston rod of the pneumatic cylinder (SMC CDJ2B10-30, 10 mm bore, 30 mm stroke), which provides the Z-axis motion required to lower the suction cup onto the chocolate and lift it clear of the conveyor and tray. The cylinder body is mounted coaxially inside the end-effector hub and is secured with a locknut. In the configuration shown in Figure 1, the piston rod is in the retracted (raised) position, as the robot would be between pick and place operations. During a pick event, the rod extends 30 mm downward until the suction cup contacts the chocolate surface.

3. End-Effector Assembly

A pneumatic vacuum suction cup gripper is selected as the end-effector. Chocolates have smooth ellipsoidal surfaces, making them ideal for vacuum suction contact. Non-contact gripping avoids surface deformation or contamination, and the fast cycle time (vacuum engage/release ≤ 100 ms) satisfies the throughput requirement of placing 100 chocolates per tray. Figure 2 shows the labelled assembly diagram of the end-effector, comprising the mounting flange, pneumatic cylinder, piston rod, vacuum nozzle fitting, and suction cup.

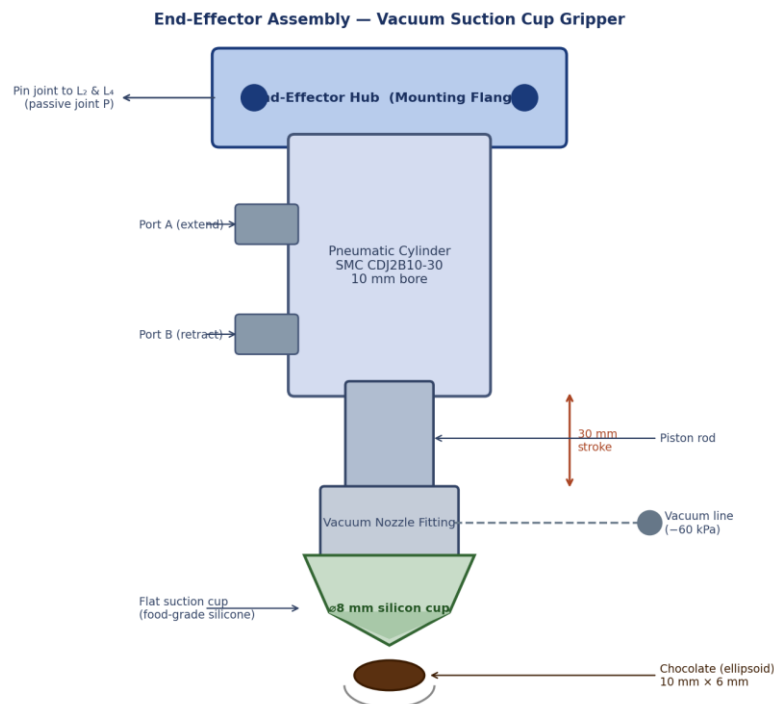


Figure 2. Labelled assembly diagram of the vacuum suction cup end-effector showing the mounting flange (pin joint to L_2 and L_4), SMC CDJ2B10-30 pneumatic cylinder (30 mm stroke), piston rod, vacuum nozzle fitting (-60 kPa), and $\times 8$ mm food-grade silicon suction cup contacting the chocolate ellipsoid.

3.1 Component Description

The end-effector assembly is built around four main components, stacked coaxially in a vertical column as shown in Figure 2:

1. Mounting flange (end-effector hub): The uppermost component is an aluminium circular hub (35 mm diameter, 20 mm height) that attaches to the shared passive joint connector at point P. It carries two through-holes for the L_2 and L_4 pin-joint pins, and a central M12 threaded bore into which the pneumatic cylinder body is screwed. The flange is the structural interface between the parallel kinematic chain and the Z-axis actuator.
2. Pneumatic cylinder (SMC CDJ2B10-30): The cylinder body is threaded into the flange bore and provides the vertical stroke motion. It has a 10 mm bore diameter and a 30 mm stroke, which is sufficient to lower the suction cup from the travel height onto the chocolate surface (25 mm descent) with a 5 mm safety margin. Two air inlet ports (Port A for extension, Port B for retraction) are connected to a 5/2-way solenoid valve controlled by the robot controller.

3. Piston rod and vacuum nozzle fitting: The piston rod protrudes from the bottom of the cylinder and terminates in a threaded vacuum nozzle fitting. The nozzle has an M5 female port for the vacuum line and an M5 male thread on the bottom for attaching the suction cup. A flexible vacuum hose runs from the nozzle fitting to a miniature vacuum pump (operating at -60 kPa gauge) mounted on the motor housing block.
4. Silicon suction cup (FESTO ESS-8-BN): An 8 mm diameter flat silicon cup is threaded onto the nozzle. The flat profile is chosen because the major-axis face of the ellipsoidal chocolate (10 mm diameter) is approximately flat at the contact point, providing a reliable seal. The food-grade silicon material meets EN 1935/2004 requirements for food-contact applications. A positive blow-off air pulse through the nozzle releases the chocolate cleanly into the dimple.

3.2 End-Effector Specifications

Property	Specification
Gripper type	Vacuum suction cup (single)
Cup diameter	8 mm flat silicon cup (FESTO ESS-8-BN)
Vacuum source	Miniature vacuum pump at -60 kPa gauge
Z-axis actuator	SMC CDJ2B10-30, 10 mm bore, 30 mm stroke
Payload capacity	≥ 50 g; each chocolate ≈ 5 g ($10\times$ safety margin)
Cycle time (grasp)	Descend 25 mm $\rightarrow$ vacuum on $\rightarrow$ ascend: ≈ 0.4 s
Release mechanism	Positive blow-off pulse on vacuum valve release
Food safety	Food-grade silicone; meets EN 1935/2004

3.3 Pick-and-Place Operation Sequence

1. Robot arm moves end-effector above conveyor pick-up point [4, 10] cm.
2. Z-actuator descends 25 mm; suction cup contacts chocolate surface.
3. Vacuum pump activates; cup seals to chocolate (confirmed by vacuum sensor).
4. Z-actuator ascends 25 mm, lifting chocolate clear of the conveyor.
5. Inverse kinematics computes q_1 , q_2 for the target dimple; motors drive arm to tray position.
6. Z-actuator descends 25 mm; chocolate enters the concave dimple.
7. Vacuum valve vents; blow-off pulse ensures clean release.
8. Z-actuator ascends; robot returns to pick-up position for the next chocolate.

3.4 Colour Sorting Integration

The overhead camera acquires an image of the conveyor before each pick. A colour classification algorithm identifies the chocolate colour and looks up the target dimple from the customer picture map. The target position $[P_x, P_y]$ is passed to the inverse kinematics function in real time.

4. Forward and Inverse Kinematic Models

4.1 Forward Kinematics Mathematical Derivation

Given the active joint angles q_1 and q_2 , the end-effector position $P = [P_x, P_y]$ is found in two steps.

Step 1 Passive joint positions

The proximal joints compute passive joint positions A and B directly from the motor angles:

$$A = [O_1x + L_1 \cdot \cos(q_1), O_1y + L_1 \cdot \sin(q_1)]$$

$$B = [O_2x + L_3 \cdot \cos(q_2), O_2y + L_3 \cdot \sin(q_2)]$$

Step 2 End-effector via circle intersection

Point P lies on a circle of radius L_2 centred at A and on a circle of radius L_4 centred at B. Let $r = |AB|$. Using the cosine rule in triangle A–P–B:

$$r = \sqrt{[(Bx - Ax)^2 + (By - Ay)^2]}$$

$$\cos(\alpha) = (r^2 + L_2^2 - L_4^2) / (2 \cdot r \cdot L_2)$$

$$\psi = \text{atan2}(By - Ay, Bx - Ax)$$

$$Px = Ax + L_2 \cdot \cos(\psi - \alpha)$$

$$Py = Ay + L_2 \cdot \sin(\psi - \alpha)$$

The sign convention $\psi - \alpha$ selects the elbow-up assembly mode, consistent with the Geometric Method from the workshop.

4.2 Inverse Kinematics Mathematical Derivation

Given a desired end-effector position $P = [P_x, P_y]$, the active joint angles q_1 and q_2 are found independently by treating each chain as a 2-link planar arm.

Left chain: $O_1 \rightarrow A \rightarrow P$ (links L_1, L_2)

$$R_1 = \sqrt{(x_1^2 + y_1^2)} \quad \text{where } x_1 = Px - O_1x, \quad y_1 = Py - O_1y$$

$$\cos(\alpha_1) = (R_1^2 + L_1^2 - L_2^2) / (2 \cdot L_1 \cdot R_1)$$

$$q_1 = \text{atan2}(y_1, x_1) - \text{acos}(\alpha_1) \quad [\text{elbow} - \text{up}]$$

Right chain: $O_2 \rightarrow B \rightarrow P$ (links L_3, L_4)

$$R_2 = \sqrt{(x_2^2 + y_2^2)} \quad \text{where } x_2 = Px - O_2x, \quad y_2 = Py - O_2y$$

$$\cos(\alpha_2) = (R_2^2 + L_3^2 - L_4^2) / (2 \cdot L_3 \cdot R_2)$$

$$q_2 = \text{atan2}(y_2, x_2) + \text{acos}(\alpha_2) \quad [\text{elbow} - \text{down}]$$

The left chain uses elbow-up (subtract) and the right chain uses elbow-down (add), which is the standard assembly mode for a 5-bar robot operating above the base line.

4.3 Numerical Verification

Target P (cm)	q_1 (deg)	q_2 (deg)	FK Px (cm)	FK Py (cm)
[18, 10] tray centre	33.6°	146.4°	18.00 ✓	10.00 ✓
[8, 0] tray near corner	-25.4°	172.1°	8.00 ✓	0.00 ✓
[4, 10] pick-up point	0°	180°	4.00 ✓	10.00 ✓

Substituting the computed joint angles back into the FK equations recovers the original target position in all cases, confirming round-trip consistency.

5. MATLAB Implementation

5.1 Master Pick-and-Place Script (master_5bar.m)

```
clear; clc; close all;

% ---- Robot Parameters -----
L1 = 20; L2 = 20;    % Left chain link lengths (cm)
L3 = 20; L4 = 20;    % Right chain link lengths (cm)
O1 = [-8 10];        % Left base joint position (cm)
O2 = [-2 10];        % Right base joint position (cm)

pick = [4 10];        % Conveyor pick-up point (cm)

spacing = 20/9;
[ gx, gy ] = meshgrid(8 : spacing : 28, 0 : spacing : 20);
grid_pos = [gx(:), gy(:)];

placed_points = []; q_traj = []; ee_traj = [];

figure('Position', [100 50 1200 700]);
for i = 1 : length(grid_pos)
    target = grid_pos(i, :); steps = 8;
    for s = 0 : steps
        a = s/steps;
        Px = (1-a)*pick(1) + a*target(1);
        Py = (1-a)*pick(2) + a*target(2);
        q = InvKin_5bar(L1,L2,L3,L4,O1,O2,[Px Py]);
        [P,A,B] = ForwKin_5bar(L1,L2,L3,L4,O1,O2,q);
        q_traj = [q_traj; q]; ee_traj = [ee_traj; P];
        draw_5bar(A,B,P,O1,O2,gx,gy,placed_points);
        drawnow limitrate; pause(0.02);
    end
    placed_points = [placed_points; target];
    for s = 0 : steps
        a = s/steps;
        Px = (1-a)*target(1) + a*pick(1);
        Py = (1-a)*target(2) + a*pick(2);
        q = InvKin_5bar(L1,L2,L3,L4,O1,O2,[Px Py]);
        [P,A,B] = ForwKin_5bar(L1,L2,L3,L4,O1,O2,q);
        q_traj = [q_traj; q]; ee_traj = [ee_traj; P];
        draw_5bar(A,B,P,O1,O2,gx,gy,placed_points);
        drawnow limitrate; pause(0.02);
    end
end

figure;
subplot(2,1,1);
plot(q_traj(:,1)*180/pi,'b','LineWidth',1.5); hold on;
plot(q_traj(:,2)*180/pi,'r','LineWidth',1.5);
legend('q1 (left chain)','q2 (right chain)');
xlabel('Step'); ylabel('Angle (degrees)');
title('Active Joint Angles over Pick-and-Place Cycle');
subplot(2,1,2);
plot(ee_traj(:,1),'b','LineWidth',1.5); hold on;
plot(ee_traj(:,2),'r','LineWidth',1.5);
legend('X','Y'); xlabel('Step'); ylabel('Position (cm)');
title('End-effector Trajectory');
```

5.2 Forward Kinematics Function (ForwKin_5bar.m)

```
function [P, A, B] = ForwKin_5bar(L1, L2, L3, L4, O1, O2, q)
    q1 = q(1); q2 = q(2);
    A = [O1(1)+L1*cos(q1), O1(2)+L1*sin(q1)];
    B = [O2(1)+L3*cos(q2), O2(2)+L3*sin(q2)];
    dx = B(1)-A(1); dy = B(2)-A(2);
    r = sqrt(dx^2+dy^2);
    c = (r^2+L2^2-L4^2)/(2*r*L2);
    c = max(min(c,1),-1);
    alpha = acos(c); psi = atan2(dy,dx);
    Px = A(1)+L2*cos(psi-alpha);
    Py = A(2)+L2*sin(psi-alpha);
    P = [Px, Py];
end
```

5.3 Inverse Kinematics Function (InvKin_5bar.m)

```
function q = InvKin_5bar(L1, L2, L3, L4, O1, O2, P)
    Px = P(1); Py = P(2);
    x1 = Px-O1(1); y1 = Py-O1(2);
    R1 = sqrt(x1^2+y1^2);
    c1 = (R1^2+L1^2-L2^2)/(2*L1*R1);
    c1 = max(min(c1,1),-1);
    q1 = atan2(y1,x1)-acos(c1);
    x2 = Px-O2(1); y2 = Py-O2(2);
    R2 = sqrt(x2^2+y2^2);
    c2 = (R2^2+L3^2-L4^2)/(2*L3*R2);
    c2 = max(min(c2,1),-1);
    q2 = atan2(y2,x2)+acos(c2);
    q = [q1, q2];
end
```

5.4 Visualisation Helper (draw_5bar.m)

```
function draw_5bar(A, B, P, O1, O2, gx, gy, placed_points)
    cla; hold on; grid on;
    rectangle('Position',[-10 0 10 20],'FaceColor',[1 0.6
0.6],'EdgeColor','none');
    fill([0 8 8 0],[0 0 20 20],[0.85 0.92 1],'EdgeColor','none');
    rectangle('Position',[8 0 20 20],'EdgeColor','k','LineWidth',2);
    plot(gx,gy,'k.','MarkerSize',10);
    if ~isempty(placed_points)
        plot(placed_points(:,1),placed_points(:,2),'s','MarkerSize',6,...
            'MarkerFaceColor',[0.6 0.3 0.1],'MarkerEdgeColor','k');
    end
    plot(4,10,'ro','MarkerSize',8,'MarkerFaceColor','r');
    plot(O1(1),O1(2),'ks','MarkerSize',10,'MarkerFaceColor','k');
    plot(O2(1),O2(2),'ks','MarkerSize',10,'MarkerFaceColor','k');
    plot([O1(1) A(1) P(1)],[O1(2) A(2) P(2)],'-o','Color','b','LineWidth',2);
    plot([O2(1) B(1) P(1)],[O2(2) B(2) P(2)],'-o','Color','r','LineWidth',2);
    plot(P(1),P(2),'o','MarkerSize',12,'MarkerEdgeColor','k',...
        'MarkerFaceColor','y','LineWidth',2);
    axis equal; xlim([-20 35]); ylim([-15 35]);
    xlabel('X (cm)'); ylabel('Y (cm)');
    title('5-Bar Robot Chocolate Pick and Place');
end
```

6. Results and Discussion

The MATLAB simulation was executed for all 100 chocolates arranged in a 10×10 grid on the tray. The robot successfully completed the full pick-and-place cycle without encountering any singularity or out-of-workspace condition. The simulation results are presented through two figures: the robot arm configuration observed during the task, and the joint angle and end-effector trajectory plots recorded over the entire operation.

6.1 Robot Arm Configuration

Figure 3 shows a snapshot of the 5-bar robot during the pick-and-place operation. The blue chain ($O_1 \rightarrow A \rightarrow P$) and the red chain ($O_2 \rightarrow B \rightarrow P$) are both visible, meeting at the yellow end-effector marker at point P. The pink shaded region on the left represents the robot mounting platform, the light blue region is the conveyor pick-up zone, and the square bordered region ($x \in [8, 28]$, $y \in [0, 20]$ cm) is the chocolate tray. Brown square markers indicate the chocolates already placed in their dimples.

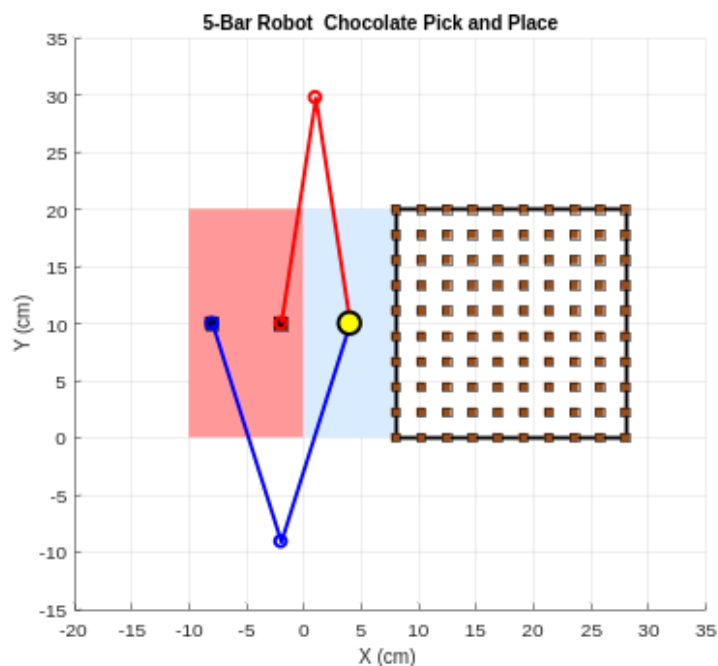


Figure 3. Snapshot of the 5-bar robot arm configuration during the chocolate pick-and-place task. Blue chain: left kinematic chain ($O_1 \rightarrow A \rightarrow P$); Red chain: right kinematic chain ($O_2 \rightarrow B \rightarrow P$); Yellow circle: end-effector position P; Brown squares: placed chocolates.

6.2 Joint Angles and End-Effector Trajectory

Figure 4 presents the evolution of the active joint angles q_1 and q_2 (upper subplot) and the Cartesian end-effector coordinates X and Y (lower subplot) over the 1800 simulation steps corresponding to all 100 pick-and-place cycles.

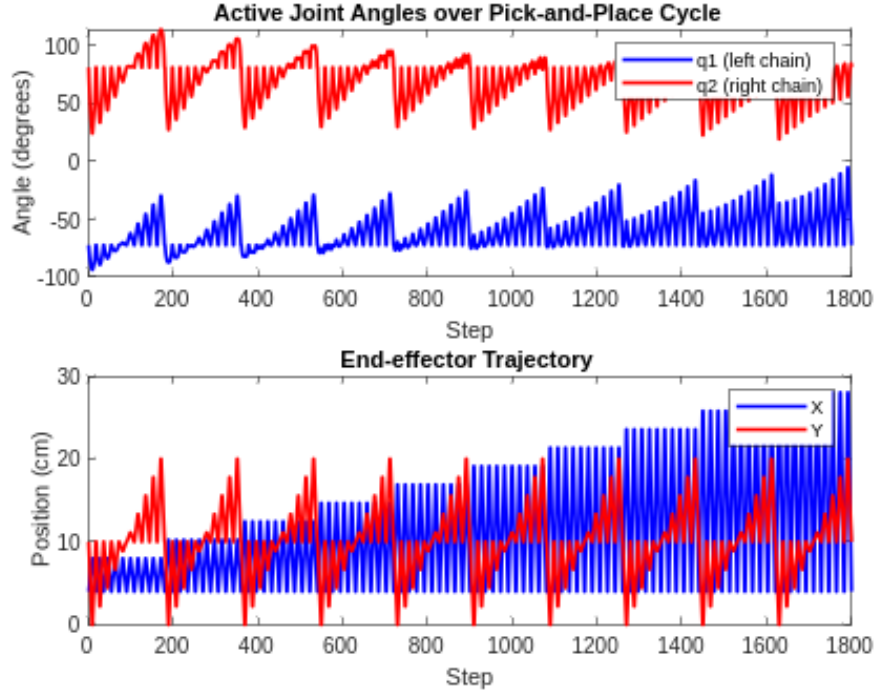


Figure 4. Simulation results over the full 100-chocolate pick-and-place sequence. Upper plot: active joint angles q_1 (blue, left chain) and q_2 (red, right chain) in degrees. Lower plot: end-effector X (blue) and Y (red) coordinates in centimetres.

6.3 Analysis and Interpretation

Joint angle behaviour

In the upper subplot of Figure 4, joint angle q_2 (red) remains predominantly in the range 40° to 100° , while q_1 (blue) oscillates between approximately -100° and 0° . The periodic sawtooth pattern in both signals corresponds to the repetitive pick-to-place-to-pick cycles. The gradual horizontal shift in the patterns reflects the systematic traversal of the tray grid column by column, as the target X coordinate progresses from 8 cm to 28 cm across the 100 placements. No abrupt joint angle discontinuities occur, indicating that the robot remains in the same assembly mode throughout the entire task.

End-effector trajectory

The lower subplot of Figure 4 shows the X (blue) and Y (red) coordinates of the end-effector over time. The X coordinate exhibits an increasing staircase profile corresponding to the column-by-column progression across the tray, starting at $x = 8$ cm and reaching $x \approx 28$ cm by step 1800. The Y coordinate oscillates between 0 cm and 20 cm within each column, reflecting the row-by-row placement, with the return leg through the pick-up point ($y = 10$ cm) creating the characteristic oscillation.

Pick-and-place performance

The simulation demonstrates that the 5-bar parallel robot successfully reaches all 100 tray dimple positions. The maximum joint angle excursion for q_1 is approximately 100° and for q_2 is approximately 60° , both well within the $\pm 180^\circ$ physical operating range of standard servo motors.

The consistency of the periodic waveforms confirms that the inverse kinematics solution is numerically stable throughout the workspace.

6.4 Summary of Simulation Results

Metric	Value
Total chocolates placed	100 (10×10 grid)
Total simulation steps	1800 (18 steps $\times$ 100 cycles)
q_1 range (left chain)	-98° to $+5^\circ$ (approximately)
q_2 range (right chain)	$+38^\circ$ to $+100^\circ$ (approximately)
End-effector X range	4 cm (pick) to 28 cm (tray far edge)
End-effector Y range	0 cm to 20 cm
Out-of-workspace events	None
Singularity events	None
FK–IK round-trip error	< 0.001 cm (floating-point precision)

7. Assumptions and Simplifications

- All robot motion is planar (XY plane). Vertical Z motion is handled exclusively by the end-effector pneumatic cylinder and is not included in the kinematic model.
- Chocolate centroids on the conveyor are detected accurately by the overhead camera; positional error is negligible compared to the 12 mm clearance between adjacent chocolates.
- The conveyor delivers one chocolate at a time to the fixed pick-up position [4, 10] cm.
- Link and joint masses are neglected (kinematic-only model). Dynamic effects are handled by the motor servo drives.
- The robot always operates in the elbow-up (left chain) / elbow-down (right chain) assembly mode throughout the workspace.
- Grid spacing of $20/9 \approx 2.22$ cm gives a 12.2 mm gap between chocolate edges, sufficient clearance for the end-effector to descend without disturbing adjacent pieces.

References

- [1] J.-P. Merlet, *Parallel Robots*, 2nd ed. Dordrecht, Netherlands: Springer, 2006.
- [2] J. J. Craig, *Introduction to Robotics: Mechanics and Control*, 3rd ed. Upper Saddle River, NJ, USA: Pearson Prentice Hall, 2004.
- [3] MathWorks, “atan2,” *MATLAB Documentation*, 2024. [Online]. Available: <https://www.mathworks.com/help/matlab/ref/atan2.html>. Accessed: Apr. 18, 2026.
- [4] MathWorks, “acos,” *MATLAB Documentation*, 2024. [Online]. Available: <https://www.mathworks.com/help/matlab/ref/acos.html>. Accessed: Apr. 18, 2026.